Q1 2021 (01 Sep Shift 2)

Let $f(x)=x^{6}+2x^{4}+x^{3}+2x+3, x\in R$. Then the natural number n for which $\lim_{\mathrm{x}\rightarrow 1}\frac{\mathrm{x}^{\mathrm{n}}\mathrm{f}(1)-\mathrm{f}(\mathrm{x})}{\mathrm{x}-1}=44$ is .

Q2 2021 (01 Sep Shift 2)

Let $f:\mathbf{R}\rightarrow\mathbf{R}$ be a continuous function. Then

$\lim_{x\rightarrow\frac{\pi}{4}}\frac{\frac{\pi}{4}\int_{2}^{\sec^{2}x}f(x)dx}{x^{2}-\frac{\pi^{2}}{16}}$ is equal to :

(1) $f(2)$

(2) $2f(2)$

(3) $2f(\sqrt{2})$

(4) $4f(2)$

Q3 2021 (31 Aug Shift 2)

If $\alpha=\lim_{\mathrm{x}\rightarrow\pi/4}\frac{\tan^{3}\mathrm{x}-\tan\mathrm{x}}{\cos\left(\mathrm{x}+\frac{\pi}{4}\right)}$ and $\beta=\lim_{\mathrm{x}\rightarrow 0}(\cos\mathrm{x})^{\cot\mathrm{x}}$ are

the roots of the equation, $ax^{2}+bx-4=0$, then the ordered pair (a,b) is :

(1) $(1,-3)$

(2) $(-1,3)$

(3) $(-1,-3)$

(4) $(1,3)$

Q4 2021 (31 Aug Shift 1)

$\lim_{x\rightarrow 0}\frac{\sin^{2}\left(\pi\cos^{4}x\right)}{x^{4}}$ is equal to :

(1) π^{2}

(2) $2\pi^{2}$

(3) $4\pi^{2}$

(4) 4π

Q5 2021 (27 Aug Shift 2)

If $\lim_{x \rightarrow \infty}\left(\sqrt{x^{2}-x+1}-a x\right)=b$, then the ordered pair (a, b) is:

(1) $\left(1, \frac{1}{2}\right)$

(2) $\left(1,-\frac{1}{2}\right)$

(3) $\left(-1, \frac{1}{2}\right)$

(4) $\left(-1,-\frac{1}{2}\right)$

Q6 2021 (27 Aug Shift 1)

If α, β are the distinct roots of $x^{2}+b x+c=0$,

then $\lim_{x \rightarrow \beta} \frac{e^{2\left(x^{2}+b x+c\right)}-1-2\left(x^{2}+b x+c\right)}{(x-\beta)^{2}}$ is equal to:

(1) $b^{2}+4 c$

(2) $2\left(b^{2}+4 c\right)$

(3) $2\left(\mathrm{b}^{2}-4 \mathrm{c}\right)$

(4) $b^{2}-4 c$

Q7 2021 (26 Aug Shift 2)

$\lim_{x \rightarrow 2}\left(\sum_{n=1}^{9} \frac{x}{n(n+1) x^{2}+2(2 n+1) x+4}\right)$ is equal to :

(1) $\frac{9}{44}$

(2) $\frac{5}{24}$

(3) $\frac{1}{5}$

(4) $\frac{7}{36}$

Answer Key

Q1 (7)
Q2 (2)
Q3 (4)
Q4 (3)
Q5 (2)
Q6 (3)
Q7 (1)

Q1 (7)

$$f(n)=x^6+2x^4+x^3+2x+3$$
$$\lim_{x\to1}\frac{x^nf(1)-f(x)}{x-1}=44$$
$$\lim_{x\to1}\frac{9x^n-\left(x^6+2x^4+x^3+2x+3\right)}{x-1}=44$$
$$\lim_{x\to1}\frac{9nx^{n-1}-\left(6x^5+8x^3+3x^2+2\right)}{1}=44$$
$$\Rightarrow 9n-(19)=44$$
$$\Rightarrow 9n=63$$
$$\Rightarrow n=7$$

Q2 (2)

$$\lim_{x\to\frac{\pi}{4}}\frac{\frac{\pi}{4}\int_2^{\sec^2x}\mathrm{f(x)dx}}{\mathrm{x}^2-\frac{\pi^2}{16}}$$
$$\lim_{x\to\frac{\pi}{4}}\frac{\pi}{4}\cdot\frac{\left[f\left(\sec^2x\right)\cdot2\sec x\cdot\sec x\tan x\right]}{2x}$$
$$\lim_{x\to\frac{\pi}{4}}\frac{\pi}{4}f\left(\sec^2x\right)\cdot\sec^3x\cdot\frac{\sin x}{x}$$
$$\frac{\pi}{4}f(2)\cdot(\sqrt{2})^3\cdot\frac{1}{\sqrt{2}}\times\frac{4}{\pi}$$
$$\Rightarrow 2\mathrm{f}(2)$$

Q3 (4)

$$\alpha=\lim_{x\to\frac{\pi}{4}}\frac{\tan^3x-\tan x}{\cos\left(x+\frac{\pi}{4}\right)};\frac{0}{0}\text{ form}$$

Using L Hopital rule

$$\alpha=\lim_{x\to\frac{\pi}{4}}\frac{3\tan^2x\sec^2x-\sec^2x}{-\sin\left(x+\frac{\pi}{4}\right)}$$
$$\Rightarrow\alpha=-4$$
$$\beta=\lim_{x\to0}(\cos x)^{\cot x}=e^{\lim_{x\to0}\frac{(\cos x-1)}{\tan x}}$$
$$\beta=\mathrm{e}\quad\lim_{x\to0}\frac{-(1-\cos x)}{x^2},\frac{x^2}{\left(\frac{\tan x}{x}\right)^x}$$
$$\beta=\mathrm{e}^{\lim_{x\to0}\left(\frac{-1}{2}\right)\cdot\frac{x}{1}}=\mathrm{e}^0\Rightarrow\beta=1$$
$$\alpha=-4;\beta=1$$

If $ax^2 + bx - 4 = 0$ are the roots then

$16a - 4b - 4 = 0 \& a + b - 4 = 0$

$\Rightarrow a = 1 \& \quad b = 3$

Q4 (3)

$\lim_{x \to 0} \frac{\sin^2(\pi \cos^4 x)}{x^4}$

$\lim_{x \to 0} \frac{1-\cos(2\pi \cos^4 x)}{2x^4}$

$\lim_{x \to 0} \frac{1-\cos(2\pi - 2\pi \cos^4 x)}{[2\pi(1-\cos^4 x)]^2} 4\pi^2 \cdot \frac{\sin^4 x}{2x^4} \left(1+\cos^2 x\right)^2$

$= \frac{1}{2} \cdot 4\pi^2 \cdot \frac{1}{2}(2)^2 = 4\pi^2$

Q5 (2)

$\lim_{x \to \infty} \left(\sqrt{x^2 - x + 1}\right) - ax = b \quad (\infty - \infty)$

$\Rightarrow a > 0$

Now, $\lim_{x \to \infty} \frac{(x^2 - x + 1 - a^2x^2)}{\sqrt{x^2 - x + 1} + ax} = b$

$\Rightarrow \lim_{x \to \infty} \frac{(1-a^2)x^2 - x + 1}{\sqrt{x^2 - x + 1} + ax} = b$

$\Rightarrow \lim_{x \to \infty} \frac{(1-a^2)x^2 - x + 1}{x\left(\sqrt{1 - \frac{1}{x} + \frac{1}{x^2}} + a\right)} = b$

$\Rightarrow 1 - a^2 = 0 \Rightarrow a = 1$

Now, $\lim_{x \to \infty} \frac{-x+1}{x\left(\sqrt{1 - \frac{1}{x} + \frac{1}{x^2}} + a\right)} = b$

$\Rightarrow \frac{-1}{1+a} = b \Rightarrow b = -\frac{1}{2}$

$(a, b) = \left(1, -\frac{1}{2}\right)$

Q6 (3)

$\lim_{x \to \beta} \frac{e^{2\left(x^2+bx+c\right)} - 1 - 2\left(x^2+bx+c\right)}{(x-\beta)^2}$

$\Rightarrow \lim_{x \to \beta} \frac{1\left(1 + \frac{2\left(x^2+bx+c\right)}{1!} + \frac{2^2\left(x^2+bx+c\right)^2}{2!} + \ldots\right) - 1 - 2\left(x^2+bx+c\right)}{(x-\beta)^2}$

$\Rightarrow \lim_{x \to \beta} \frac{2\left(x^2+bx+1\right)^2}{(x-\beta)^2}$

$\Rightarrow \lim_{x \to \beta} \frac{2(x-\alpha)^2(x-\beta)^2}{(x-\beta)^2}$

$\Rightarrow 2(\beta-\alpha)^2 = 2\left(b^2-4c\right)$

Q7 (1)

$\mathrm{S} = \lim_{x \to 2} \sum_{\mathrm{n}=1}^{9} \frac{\mathrm{x}}{\mathrm{n}(\mathrm{n}+1)\mathrm{x}^2+2(2\mathrm{n}+1)\mathrm{x}+4}$

$\mathrm{S} = \sum_{\mathrm{n}=1}^{9} \frac{2}{4(\mathrm{n}^2+3\mathrm{n}+2)} = \frac{1}{2} \sum_{\mathrm{n}=1}^{9} \left(\frac{1}{\mathrm{n}+1} - \frac{1}{\mathrm{n}+2}\right)$

$\mathrm{S} = \frac{1}{2}\left(\frac{1}{2} - \frac{1}{11}\right) = \frac{9}{44}$